MILITARY
SPECIFICATION

MIL-M-13786(SigC)
15 November 1954

MOTORS, FRACTIONAL HORSEPOWER, DIRECT CURRENT AND UNIVERSAL
(For communication and other electronic and special military applications)

1. SCOPE

1.1 This specification covers direct-current and universal, fractional horsepower motors, having the specific environmental characteristics required for communication and other electronic and special military applications, and capable of withstanding extreme service and environmental conditions such as shock, high temperature, low temperature, vibration, humidity, fungus, etc. A fractional horsepower motor is a motor built in a frame smaller than that of a motor having a continuous rating of one horsepower, open construction, at 1700 to 1800 revolutions per minute (i.e., the outside diameter of the frame should be less than 10 inches). This general specification is applicable as a procurement document when used with a complementary document (e. g., specification sheets, service drawings, or other documents) giving detail requirements and applicable additions or exceptions to the general requirements and tests specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the isue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

FF-B-171	Bearings; Ball.
NN-B-621	Boxes, Wood, Nailed and Lock Corner.
NN-B-631	Boxes; Wood, Wirebound (For Domestic Shipment).
QQ-S-571	Solder, Soft (Tin, Tin-Lead and Lead-Silver).
UU-T-111	Tape, Paper, Gummed (Kraft).
LLL-F-291	Fiberboard; Corrugated, Single-Face (Flexible).
LLL-B-631	Boxes; Fiber, Corrugated (For Domestic Shipment).
PPP-B-601	Boxes; Wood-Cleated-Plywood.

MILITARY

MIL-C-5	Capacitors, Mica-Dielectric, Fixed.
MIL-P-14	Plastic-Materials, Molding, and Plastic-Parts, Molded; Thermosetting.
MIL-C-25	Capacitors, Fixed, Paper-Dielectric, Direct-Current (Hermetically Sealed in Metallic Cases).
MIL-P-79	Plastic-Materials, Laminated, Thermosetting Rods and Tubes.
JAN-C-91	Capacitors, Paper-Dielectric, Fixed (Nonmetallic Cases).

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JAN-P-106	Packaging and Packing For Overseas Shipment - Boxes, Wood, Nailed, (For Weight Of Contents Not In Excess Of 1000 Pounds).
JAN-P-108	Packaging and Packing For Overseas Shipment Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116	Preservation, Methods Of.
JAN-B-121	Barrier Materials, Greaseproof.
MIL-C-124	Containers, (Cans, Pails and Drums), Metal (For Other Than Subsistance Items).
JAN-P-127	Packaging and Packing For Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant.
MIL-B-131	Barrier Material, Water-Vapor-Proof, Flexible.
JAN-P-133	Packaging and Packing For Overseas Shipment - Boxes, Set-Up, Paperboard.
JAN-T-152	Treatment, Moisture And Fungus Resistant, Of Communications, Electronic, And Associated Electrical Equipment, General Process For.
JAN-E-251	Electrical-Equipment: Rotation, Connections, And Terminal Markings For.
JAN-W-583	Wire, Magnet
MIL-S-901	Shockproof Equipment, Class HI (High-Impact), Shipload.
MIL-P-997	Plastic-Material, Laminated, Thermosetting Electrical-Insulating: Sheets, Glass Cloth, Silicone Resin.
MIL-V-1137	Varnish, Electrical-Insulating (For Electro-Motive Equipment).
MIL-P-3115	Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic-Resin.
MIL-G-3278	Grease; Aircraft And Instruments (For Low And High Temperatures).
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-O-5606	Oil; Hydraulic, Aircraft, Petroleum Base.
MIL-L-6085	Lubricating Oil; Aircraft Instrument, Low Volatility
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, And Nickel Alloys.
MIL-T-10513	Tropicalization Of Signal Corps Equipment.
MIL-S-11748	Suppression requirements for Electricaland electronic equipment (except internal combustion engine-driven equipment)
MIL-P-15037	Plastic-Material, Laminated Thermosetting, Sheets, Glass-Cloth, Melamine-Resin.
MIL-P-15047	Plastic-Material, Laminated Thermosetting, Sheets, Nylon Fabric Base, Phenolic-Resin.

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U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Single And Double-Faced, For General Packing Purposes.
72-53	Finishes (For Ground Signal Equipment).

U. S. AIR FORCE

7201	Test Procedure: Shock, Medium Impact, Variable - Duration, Method and Apparatus For.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129	Marking of Shipments.
MIL-STD-202	Test Methods For Electronic And Electric Component Parts.

SPECIFICATION SHEETS

(Specification sheets for motors covered by this specification when issued, will be in supplement to this specification.)

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards For Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

3. REQUIREMENTS

3.1 Preproduction samples. -

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3.1.1 Quantities. - After award of contract, six samples of each type of motor on order shall be furnished for Government inspection and approval. Two sets of spare brushes shall be supplied for each specimen.

3.1.2 Compliance with specified requirements. -

3.1.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production and shall be accompanied by a statement to that effect. (See 3.1.2.3 for exception.)

3.1.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.1.5), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.1.2.3 When deviation from 3.1.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefore, and the manner in which it will be corrected in production of motors on contract.

3.1.3 Test data. - The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.1.4 Preproduction inspection. - Preproduction samples will be subjected by the Government to the preproduction inspection tests specified in Table I, in the order shown. After completion of Group I tests (see table I) the samples will be divided into two equal groups and subjected to the tests for their particular group.

3.1.5 Reference standards. - After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.2 Materials. -

3.2.1 Restricted materials. -

3.2.1.1 Inflammable and explosive materials. - Motors shall incorporate in their construction the smallest practicable amount of inflammable and explosive materials consistent with good commercial practice.

3.2.1.2 Corrosive materials. - Corrosive materials used in any of the processes in the manufacture of motors shall be neutralized to minimize corrosion in the completed unit. Materials used in the construction of motors shall be of a corrosion-resistant type or shall be treated to resist corrosion.

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Motors, Fractional Horsepower, Direct Current and Universal

Table I - Preproduction Inspection

Test	Requirement paragraph	Test paragraph
Test Group I (all of sample)		
Visual and mechanical inspection (external) 1/	3.2.1 to 3.2.3 incl 3.4 to 3.9.2.2 incl 3.9.6 3.9.7 3.25.1 to 3.25.6 incl 3.26 3.27	4.2.1
Terminal or lead strength	3.11	4.2.2
Speed-torque characteristics		4.2.3
Locked-Armature protection (When applicable)	3.12	4.2.4
Overspeed (When applicable)	3.13	4.2.5
Insulation Resistance	3.14	4.2.6
Dielectric Strength	3.15	4.2.7
Dynamic balance	3.17	4.2.9
Temperature rise 2/		4.2.11
Temperature cycling	3.18	4.2.12
Concentricity	3.16	4.2.1
Test Group II (1/2 of sample) 3/		
Radio-frequency interference		4.2.14
Overload	3.20	4.2.15
Low-temperature storage and operation	3.21	4.2.16
Mechanical shock	3.22	4.2.17
High-altitude operation (when applicable)		4.2.18
Moisture resistance	3.23	4.2.19
Tropicalization	3.6	4.2.20
Visual and Mechanical inspection	3.2.1 to 3.9.2.1 incl 3.9.3 to 3.9.5 incl 3.26 3.27	
Test Group III (1/2 of sample) 3/		
Vibration	3.19	4.2.13
Life		4.2.21
Corrosion (when applicable)	3.24	4.2.22
Motor Enclosure	3.25 to 3.25.11 incl	4.2.23
Visual and mechanical inspection (internal) 1/	3.9.3 to 3.9.5 incl 3.9.3 to 3.9.5 incl 3.26 3.27	4.2.1

1/ Marking will be considered defective only if it is illegible at the completion of any of the required tests.
2/ Only two specimens will be subjected to the temperature-rise test.
3/ One of the specimens which has been subjected to the temperature-rise test will be included in each of test groups II and III.

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3.2.2 Plastic. -

3.2.2.1 Laminated. - Laminated plastic sheets shall be in accordance with Specification MIL-P-997, MIL-P-3115, MIL-P-15037, or MIL-P-15047, as applicable. Laminated plastic rods and tubes shall be in accordance with Specification MIL-P-79. All tubes shall be seamless rolled.

3.2.2.2 Molded. - Molded plastic-material except where used for wearing parts of switch mechanisms shall be in accordance with Specification MIL-P-14. Where a high degree of arc resistance is required, type CMG, MME, or CMI-5 shall be used.

3.2.3 Wire. - Magnet wire shall conform to Specification JAN-W-583; however, wire having a diameter smaller than 0.0031 inch (AWG Size 40) and wire used in motors having class B or H insulation (see 6.5) shall require prior approval by the contracting officer. Lead wire shall be insulated, flexible, stranded wire.

3.2.3.1 Electrical insulation. - Motor windings which operate at a hot spot temperature (maximum ambient temperature plus temperature rise) of 105°C or less shall contain Class A insulation. Motor windings which operate at a hot-spot temperature between 105°C and 135°C shall contain Class B insulation. Motor windings which operate at a hot-spot temperature between 135°C and 200°C shall contain class H insulation. (See 6.5.)

3.2.4 Lubricants. - Lubricants supplied in motors which require relubrication shall conform to Specification MIL-G-3278 or MIL-L-6085, as applicable.

3.2.5 Capacitors. - When the styles, ratings, and characteristics are suitable, capacitors shall be in accordance with Specification MIL-C-5, MIL-C-25, or JAN-C-91.

3.3 Design and construction. -

3.3.1 Bearings. - Ball bearings, when used, shall conform to Specification FF-B-171.

3.3.1.1 Bearing mounting. - When bearings are mounted in soft metal such as zinc alloys, aluminum, or copper, a steel insert shall be provided when specified.

3.3.2 Impregnation. - Windings shall be impregnated. Impregnating materials and processes shall be such that there will be no flowing, cracking, chipping, crazing, or giving off of fumes that will interfere with motor operation. Impregnating varnish shall conform to Specification MIL-V-1137, Grade CB, unless otherwise specified by the contracting officer.

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3.4 Brazing and soldering. -

3.4.1 Brazing. - Brazing of steel, copper, copper alloys, and nickel alloys shall conform to Specification MIL-B-7883.

3.4.2 Soldering. -

3.4.2.1 Solder. - Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571. Solder shall not be used primarily for obtaining mechanical strength. All external terminals used for solder connections shall be hot-tin dipped to facilitate soldering.

3.4.2.2 Flux and cleaning agents. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.2.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. The soldering shall be so accomplished that no rosin remains on the joint or in the proximity of the joint. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operating shall be undamaged and parts fastened thereto shall not have become loosened.

3.5 Cleaning. - Metal parts after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

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resolve any differences of opinion regarding interpretation of requirements

3.6 Tropicalization. - The assembled equipment shall be treated in accordance with Specification JAN-T-152. Before being assembled into the equipment, materials and parts shall be treated in accordance with Specification MIL-T-10513.

3.7 Welding. - Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, noncritical structures. Wherever possible, the welded joint shall be such that grinding on the finished weld will be unnecessary.

3.7.1 Cleaning. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.7.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum or magnesium.

3.8 Securing of wire leads. - Field coil leads shall be firmly secured to the windings by mechanical means in addition to the solder connection. Where leads are brought from the housing, means shall be provided so that no external strain can be transmitted to the winding unless otherwise specified.

3.9 Screw threads, threaded parts, and related items. -

3.9.1 Screw threads. - Screw threads shall conform to Handbook H28 and supplement thereto. The Unified Thread Form and Series shall be class 2A or 2B fit and the American National Thread Series shall be class 2 fit. Except for holding labels or similar lightweight parts not used for structural purposes or critical applications, threads in aluminum or other soft material shall not be employed where other means of fastening are practicable and, in particular, shall not be used when frequent disassembly and reassembly may be necessary. When specified, steel inserts shall be provided for the threaded parts in soft metal which will be detached from the motor in normal operation or maintenance. When considered necessary by the Government inspector, based on his inspection of the equipment or observation of the contractor's

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production methods, the contractor shall use a torque measuring or limiting device for tightening threaded parts. When such devices are used, the values of torque shown in table II shall be applied to steel screws. When torque measuring or limiting devices are used for screws of other size, pitch, or material, the torque shall be the design value established by the contractor for the particular application.

Table II - Torque for tightening steel screws

Thread	Torque (in-lb)
6-32	9 to 12
8-32	18 to 24
10-24	29 to 34
10-32	32 to 40
12-24	42 to 56

3.9.2 Thread engagement and extension. - The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes 1/4 inch and larger, as shown in Handbook H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut.

3.9.2.1 Locking of screw-thread assemblies. - All screw-thread assemblies shall be rendered resistant to loosening under conditions of vibration.

3.9.2.2 Through-bolting. - Through-bolting shall be used wherever practicable.

3.9.3 Coating. - The exposed ends of the armature and field coils and the leads to the commutator shall be coated to protect them from damage during servicing and operation. The armature shall be insulated and coated to withstand the tests specified herein.

3.9.4 Brushes and brush holders. -

3.9.4.1 Brush material and design. - Brushes shall be of a composition which will tend to reduce commutator wear to a minimum consistent with the specified brush life at the specified operating conditions. Brushes shall be seated to the contour of the commutator so that a minimum of 75 percent of the area of the brush face makes contact with the commutator.

3.9.4.2 Spare brushes. - The number of sets of spare brushes shall be as specified. (See 6.1.1 and 6.1.2.) All spare brushes shall be formed to a radius of

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90 to 100 percent of the radius of the commutator and shall be duplicates of those supplied in the motor.

3.9.4.3 Brush holders. - Brush holders shall insure positive contact between the brush and the commutator throughout the brush life and shall be so designed that brushes may be changed without the use of special tools. When brush holders of the slotted-cartridge type are used, the design shall be such that they cannot be displaced by vibration or shifted during servicing. The design and construction shall eliminate exposed springs and shunt connections, assure positive contact between the brush terminal and the cartridge, and shall not permit rotation of the brush in the holder. All brush caps shall be insulated. When screw-type caps are used, the shunt terminal opposite the brush shall not turn in the holder.

3.9.4.4 Brush marking. - Each brush shall be permanently and legibly marked to indicate the points of 75 and 100 percent of wear. The wear points shall be so marked that there is continuous marking between the two points of wear along the length of the brush. The brush-wear marking shall be done preferably by the use of an arrowhead as shown on figure 1. The polarity and the brush grade or material shall also be marked on each brush. This marking shall be placed above the point of 100 percent of wear so that the information is still available when the brush has been worn to that point.

3.9.5 Commutators. - The surface of the commutator shall be free from burrs and other protrusions. When the segment insulation does not wear as rapidly as the segments, it shall be undercut. Undercutting shall be accomplished by removing all mica between bars to form a groove not to exceed 3/64 inch in depth.

3.9.6 Finish. - The finish shall be in accordance with specification 72-53 except that crackle or wrinkle finish shall not be used.

3.9.7 Direction of rotation. - Facing the end of the motor opposite the drive shaft extension the direction of rotation shall be determined as being either clockwise, counterclockwise, or reversible, as specified.

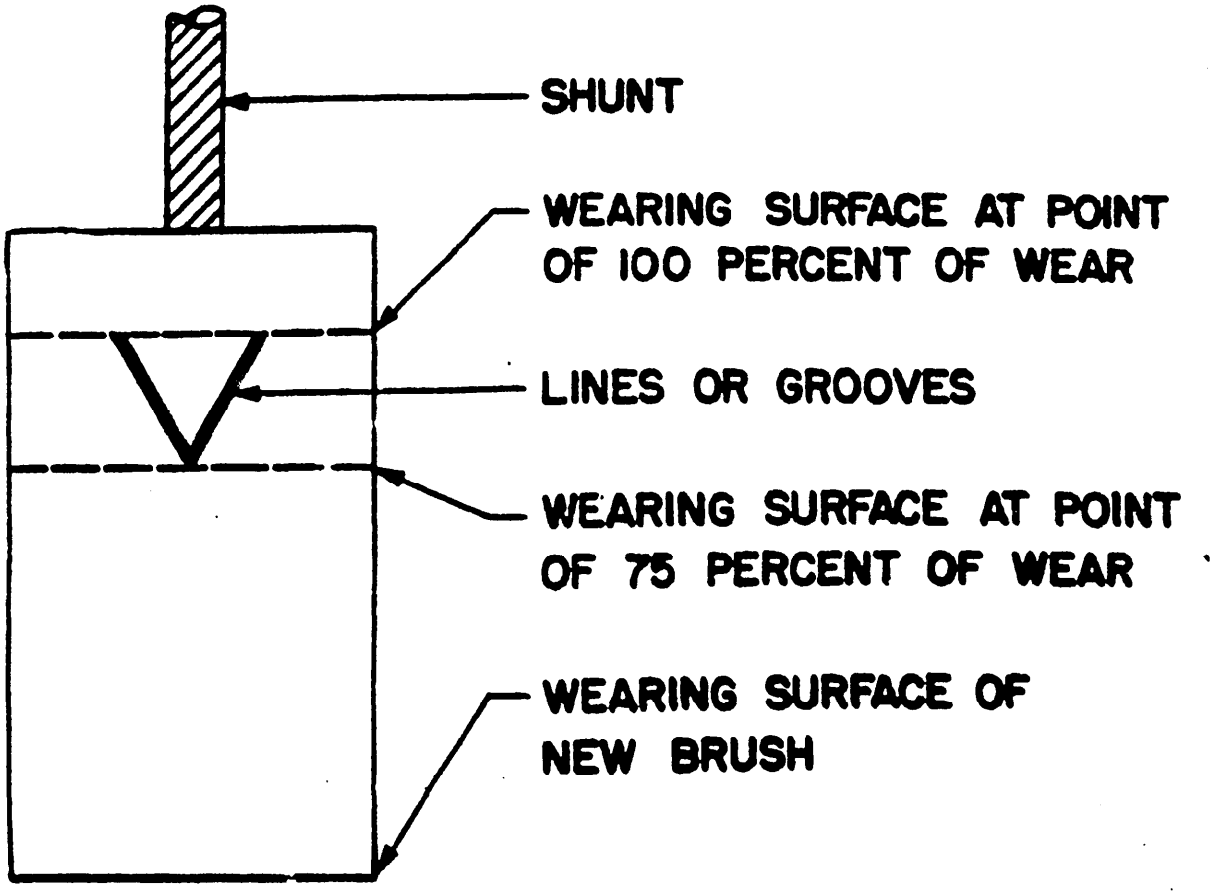
3.9.8 Run-in period. - All motors shall be subjected to a run-in period.

3.9.9 Duty cycle. - Motors shall be designed to operate at one of the following duty cycles.

3.9.9.1 Continuous. - A continuous motor is one designed for operation at full-load torque continuously at the specified maximum ambient temperature without exceeding the specified temperature rise.

3.9.9.2 Intermittent. - An intermittent motor is one designed for operation at specified regular intervals at a specified combination of full-load torque, no load, and rest.

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TYPICAL BRUSH-WEAR MARKING

FIGURE 1

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prevent the free exchange of air between the inside and the outside of the case, but not

3.10 Commutation. - During any period of operation specified herein, there shall be no sparking except fine pinpoint sparking, and after operation, there shall be no evidence of excessive wear, pitting, or other injuries to the commutator.

3.11 Terminal or lead strength. - The terminals or leads shall show no damage when tested as specified in 4.2.2.

3.12 Locked-rotor protection (applicable only to motors for which an overload-protection device is specified. - There shall be no damage to the motor or the overload protection device, and the no-load speed shall not change by more than 10 percent from the value previously determined in the no-load test (see 4.2.3.5), when motors are tested as specified in 4.2.4.

3.13 Overspeed (applicable only to motors for which external field connections are specified). - There shall be no damage to the armature, and the no-load speed shall not change by more than 10 percent from the value previously determined in the no-load test (see 4.2.3.5), when motors are tested as specified in 4.2.5.

3.14 Insulation resistance. - The motor shall have an insulation resistance of not less than 12 megohms when tested as specified in 4.2.6.

3.15 Dielectric strength. - There shall be no puncture of solid insulation, no continuing sparkover, no breakdown of dielectric material, and no damage to the insulation sufficient to cause failure of the motor, when motors are tested as specified in 4.2.7.

3.16 Concentricity. - When motors are tested as specified in 4.2.8 there shall be no rubbing or binding of the rotor.

3.17 Dynamic balance. - When motors are tested as specified in 4.2.9, the maximum total excursion shall not exceed 0.0015 inch unless otherwise specified.

3.18 Temperature cycling. - There shall be no dielectric breakdown or impairment of protective coatings, when motors are tested as specified in 4.2.12.

3.19 Vibration. - There shall be no mechanical or electrical damage, and motors shall continue to operate, when tested as specified in 4.2.13.

3.20 Overload. - When motors are tested as specified in 4.2.15, the no-load speed shall not change by more than 10 percent from the value previously determined in the no-load test (see 4.2.3.5), and the motor shall meet the requirements for dielectric strength. (See 3.15.)

3.21 Low-temperature storage and operation. - Motors shall reach 90 percent of specified full-load speed within 30 seconds for continuous motors, and within 5 seconds for intermittent motors, and shall continue to operate, when tested as specified in 4.2.16.

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3.22 Mechanical shock. -

3.22.1 Test I. - The motor shall be operative after each blow, and there shall be no binding or rubbing of parts, when tested as specified in 4.2.17.1. Momentary stopping is permissible provided the motor resumes operation after each blow.

3.22.2 Test II. - There shall be no mechanical or electrical damage, when the motor is tested as specified in 4.2.17.2.

3.23 Moisture resistance. - After the test specified in 4.2.19, the insulation resistance of the motor shall be not less than 50,000 ohms within one hour after removal from the chamber. The motor shall meet the requirements for dielectric strength (see 3.15), and full-load speed shall be as specified, when tested as specified in 4.2.19. Marking shall remain legible. After a 48-hour drying period, the insulation resistance shall not be less than 12 megohms.

3.24 Corrosion (when applicable). - There shall be no evidence of excessive corrosion, and the marking shall remain legible, when motors are tested as specified in 4.2.22.

3.25 Motor enclosure. - The motor enclosure shall conform to one or more of the following enclosures.

3.25.1 Open. - An open motor shall have ventilating openings which permit passage of external cooling air over and around the windings.

3.25.2 Protected. - A protected motor shall have ventilating openings which do not exceed 1/2 square inch in area and shall be of such shape as to prevent the passage of a rod 1/2 inch in diameter.

3.25.3 Totally enclosed. - A totally-enclosed motor shall be so enclosed as to prevent the free exchange of air between the inside and the outside of the case, but not sufficiently enclosed to be termed airtight.

3.25.4 Fan-cooled. - A fan-cooled motor shall incorporate an external fan mounted on the shaft as an aid in cooling. Drain holes shall be provided for the rapid drainage of water from the outer or fan enclosure.

3.25.5 Dripproof. - A dripproof motor is an open motor in which the ventilating openings are so constructed that drops of liquid or solid particles falling on the motor at any angle not greater than 15 degrees from the vertical cannot enter the motor, either directly or by striking and running along a horizontal or inwardly inclined surface.

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3.25.6 Splashproof. - A splashproof motor is an open motor in which the ventilating openings are so constructed that drops of liquid or solid particles falling on the motor or coming toward it in a straight line at any angle not greater than 100 degrees downward from the vertical cannot enter the motor either directly or by striking and running along a surface.

3.25.7 Oilproof. - When tested as specified in 4.2.23.1, the motor shall be operative and shall meet the requirements for dielectric strength. (See 3.15.)

3.25.8 Explosionproof. - When tested as specified in 4.2.23.2, the enclosure shall not permit ignition of the explosive vapor surrounding the motor in the test chamber by transmission of the internal explosion.

3.25.9 Watertight. - When tested as specified in 4.2.23.3, the motor shall show no evidence of water penetration, except that leakage may occur around the shaft provided water is prevented from entering the oil reservoir and provision is made for draining the motor.

3.25.10 Submersible. - When tested as specified in 4.2.23.4, the motor shall show no evidence of water penetration.

3.25.11 Dustproof. - When motors are tested as specified in 4.2.23.5, the full-load current and full-load speed shall be as specified.

3.26 Marking. - The motor shall be permanently and legibly marked with the designation "MOTOR, DIRECT CURRENT", or "MOTOR UNIVERSAL", as applicable, the specification sheet number, the manufacturer's name or symbol, and any other information specified. Terminal marking shall conform to Specification JAN-E-251.

3.27 Connections. - The electrical connections of motors shall not depend on wires, lugs, terminals, or other connectors clamped between a metallic member and an insulating material. When the maintenance of a tight connection depends on the resistance of an insulating material to compressive or shearing stress, such a connection shall be securely soldered. Electrical connections shall be made mechanically secure and electrically continuous before soldering.

3.28 Complementary documents for individual motor types. - The following requirements, if specified on complementary documents (e.g. service drawings, specifications or other documents) shall be tested in accordance with the corresponding test paragraph as indicated in section 4 of this specification.

1.	Free end play and clearance	(see 4.2.1)
2.	Dimensions and weight	(see 4.2.1)
3.	Breakdown torque	(see 4.2.10)
4.	Mechanical coupling	(see 4.2.1)

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5.	Locked-rotor torque and locked rotor-current	(see 4.2.3.1)
6.	Minimum full-load speed, full-load current, minimum full-load power input	(see 4.2.3.2)
7.	No load speed and no load current	(see 4.2.3.5)
8.	Effect of variation in line voltage	(see 4.2.3.6)
9.	Efficiency	(see 4.2.3.3)
10.	Temperature rise of the armature winding and frame	(see 4.2.11)
11.	Radio frequency interference	(see 4.2.14)
12.	High altitude operation	(see 4.2.18)
13.	Life	(see 4.2.21)

3.28.1 When requested, the manufacturer shall furnish Government complete information (manufacturers drawings, special processes, types of all material, test date), covering the manufacture and testing equipment purchased under this specification.

3.28.2 In the event of a conflict between the requirements of the general specification and the complementary document, the latter shall govern.

3.29 Workmanship. - The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of paragraphs 3.2.1 through 3.9.6 and 3.15, and all components, subassemblies, parts and final assembly shall be examined for defects of quality as listed in 4.2.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection general. - Government inspection shall consist of group A, group B and group C inspection as specified in 4.1.1 through 4.1.8 (see 6.8.1). Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.1.1 Group A inspection. - Group A inspection shall be in accordance with table III and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2 Standard test conditions. - Unless otherwise specified herein, all measurements and tests shall be made at 25 +10°C., ambient atmospheric pressure, and ambient humidity.

4.1.3 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the

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required inspection tests. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.1.4 Group A tests. - Group A tests shall consist of those specified in Table III. Group A tests shall be performed by either of the following methods or a combination of both.

(a) 100-percent inspection.

(b) Statistical sampling and inspection in accordance with Standard MIL-STD-105. The acceptable quality levels (AQL's) for visual and mechanical inspection shall be 1.5 (percent defective) for major defects and 6.5 (percent defective) for minor defects. Major and minor defects shall be as defined in Standard MIL-STD-105. For all other group A tests combined, the AQL shall be 1.5 (percent defective); all defects shall be considered major.

4.1.4.1 Lot. - An inspection lot shall be as specified in Standard MIL-STD-105.

Table III - Government group A inspection tests

Test	Requirement Paragraph	Test Paragraph
Visual and mechanical inspection (external)		4.21
Threaded parts	3.9	4.21
Finish	3.9.6	4.21
Marking	3.26	4.21
Workmanship	3.29	4.21
Speed-torque characteristics		4.2.3.1 to 4.2.3.2.2 incl & 4.2.3.5
Insulation resistance	3.14	4.2.6
Dielectric strength	3.15	4.2.7
Dynamic balance	3.17	4.2.9

4.1.5 Group B inspection. - Group B inspection shall be in accordance with table IV.

4.1.5.1 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.5.2 Sampling procedure. - Sample units shall be selected without regard to their quality by the Government inspector. Unless otherwise specified, normal inspection shall be used at the start of the contract. If the number of defective samples

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units from a lot exceeds the acceptance number specified in the sampling plan, the lot shall be rejected. Disposition of the rejected product (sample units and lots) shall be in accordance with MIL-STD-105 and 4.1.10.

4.1.5.3 Procedure in case of failure. - When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table IV - Government group B inspection tests

Test	Requirement paragraph	Test paragraph	Group B plans	Table
Terminal or lead strength	3.11	4.2.2	B-2	V
Locked-armature protection (when applicable)	3.12	4.2.4	B-2	V
Overspeed (when applicable)	3.13	4.2.5	B-2	V
Low-temperature storage and operation	3.21	4.2.16	B-2	V
Overload	3.20	4.2.15	B-2	V
High-altitude operation (when applicable)		4.2.18	B-2	V

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Table Va - Group B-2 sampling plans

Lot size /&	Reduced /# Inspection		Normal /# Inspection		Tightened /# Inspection	
	Sample size	Acceptance number	Sample size	Acceptance number	Sample size	Acceptance number
9 to 25, incl.	2	0	2	0	4	0
26 to 65, incl.	2	0	3	0	6	0
66 to 180, incl.	7	1	7	1	17	1
181 to 500, incl.	7	1	17	2	28	2
501 to 1,300, incl.	7	1	28	3	41	3
1,301 to 3,200, incl.	17	2	41	4	53	4
3,201 to 8,000, incl.	17	2	53	5	66	5
8,001 to 22,000, incl.	28	3	66	6	81	6
22,001 and over	28	3	81	7	81	6

/& For lot sizes under 9, the sample size shall be at the option of the Government but not more than 2, and the acceptance number shall be 0.

/# Normal, tightened, or reduced inspection shall be used in accordance with the following: (Part "b" of this table lists values for limits A, B, and C, referenced below.)

(1) Calculate the estimated process average from the formula 100 P/N; where P equals the number of defective units found in N sample units, and N equals the number of sample units included in the estimated process average. The Government will determine the number of consecutive sample units to be used to estimate the process average.

(2) Except as stated in (3) below, use normal inspection as long as the process average is between limits A and C.

(3) Change to tightened inspection whenever the process average becomes equal to or greater than limit A, or whenever a lot fails to pass.

(4) Return to normal inspection from tightened inspection when two lots in succession pass tightened inspection, and the process average becomes less than limit B.

(5) Change to reduced inspection whenever the process average becomes less than limit C.

(6) Return to normal inspection from reduced inspection when the process average becomes equal to or greater than limit C.

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Table Vb - Group B-2 limits of the process average

N	Limit A %	Limit B %	Limit C %	N	Limit A %	Limit B %	Limit C %
10	18.78	11.89	*	130-144	8.82	6.91	1.18
11	18.14	11.57	*	145-159	8.62	6.81	1.38
12	17.58	11.29	*	160-179	8.45	6.72	1.55
13	17.09	11.04	*	180-199	8.25	6.62	1.75
14	16.65	10.82	*	200-249	8.08	6.54	1.92
15-16	16.25	10.63	*	250-299	7.76	6.38	2.24
17-18	15.57	10.29	*	300-349	7.52	6.26	2.48
19-20	15.00	10.00	*	350-399	7.33	6.16	2.67
21-23	14.51	9.76	*	400-449	7.18	6.09	2.82
24-26	13.90	9.45	*	450-499	7.05	6.03	2.95
27-29	13.39	9.19	*	500-549	6.95	5.97	3.05
30-34	12.96	8.98	*	550-599	6.86	5.93	3.14
35-39	12.37	8.68	*	600-649	6.78	5.90	3.22
40-44	11.89	8.45	*	650-699	6.71	5.85	3.29
45-49	11.50	8.25	*	700-749	6.65	5.82	3.35
50-59	11.16	8.08	*	750-799	6.59	5.80	3.41
60-69	10.63	7.81	*	800-849	6.54	5.77	3.46
70-79	10.21	7.60	*	850-899	6.50	5.75	3.50
80-89	9.87	7.44	0.13	900-949	6.45	5.73	3.55
90-99	9.59	7.30	0.41	950-999	6.41	5.71	3.59
100-114	9.36	7.18	0.64	1000	6.38	5.69	3.62
115-129	9.06	7.03	0.94				

* Number of sample units included in estimated process average is insufficient for reduced inspection.

N = Number of sample units in estimated process average.

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4.1.6 Group C inspection. - This shall consist of group C inspection specified in table VI

4.1.7 Selection of samples. - One motor shall be selected from the first lot presented for inspection which has been subjected to and has passed the group A tests but which has not been subjected to the group B tests. When production continues for a period beyond the rated life of the sample unit under test, an additional sample unit selected from current production shall be placed on test periodically at intervals equal to the rated life of the motor, except that these intervals shall be not less than 30 days.

4.1.8 Noncompliance. - If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of corrective action taken on the process and affected portion of product. If the Government inspector does not consider that the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

Table VI - Group C inspection tests

Test	Requirement paragraph	Test paragraph
Temperature rise		4.2.11
Life		4.2.21
Visual and mechanical inspection		4.2.1
Restricted materials	3.2.1	4.2.1
Plastic	3.2.2	4.2.1
Wire	3.2.3	4.2.1
Lubricants	3.2.4	4.2.1
Bearings	3.3.1 and 3.3.1.1	4.2.1
Impregnation	3.3.2	4.2.1
Securing of wire leads	3.8	4.2.1
Threaded parts	3.9 to 3.9.2.2 incl.	4.2.1
Free end-play and clearance		4.2.1
Coating		4.2.1
Brushes and brush holders	3.9.4	4.2.1
Commutators	3.9.5	4.2.1
Dimensions and weight		4.2.1
Mechanical coupling		4.2.1
Workmanship		4.2.1

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4.1.9 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are re-subjected to and pass group A inspection (after repair of all visible damage).

4.1.10 Disposition of nonconforming product. - When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such items shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.2 Test procedures. -

4.2.1 Visual and mechanical inspection. - Parts and equipment shall be examined for defects. Classification of defects should include those listed below as major and minor.

1. Major

   a. Loose nuts and screws.
   b. Welding - scabs, blisters, abnormal pock marks and other harmful defects.
   c. Soldering - less than 1/2 turn, insufficient solder, movement of wire or solder upon pick application, insulation in terminal hole.
   d. Wiring - broken strands, damaged or frayed insulation, wire exhibits no slack.
   e. Riveting - rivet too close to the edge of the material riveted, damage to the joined parts, rivet head improperly seated.
   f. Painting - color not as specified, unsatisfactory adhesion, film thickness less than 1/2 specified thickness.
   g. Marking - missing, illegible or wrong marking.
   h. Missing or unoperative parts improperly assembled.
   i. Dimensions - incorrect dimensions of shaft and mounting.
   j. Improper brush seating.

2. Minor

   a. Insulation damaged or deteriorated, but short circuit not possible.
   b. Scratches, cuts, abrasions, small amounts of flaking, peeling or chipping.
   c. Tight wiring with no possibility of breakage.
   d. Useless holes or slots
   e. Dimensions - incorrect overall dimensions.

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4.2.2 Terminal or lead strength (see 3.11). -

4.2.2.1 Pull. - A pull of 5 pounds, unless otherwise specified, shall be applied to each lead or terminal at the point where the lead from the external circuit connects to it. Where a contacting device (e.g., cable connector, coaxial connector, or similar device) is used, the pull shall be applied to the assembled unit. The pull may be applied in any direction, and shall be increased gradually to the specified magnitude and held at that value for at least 30 seconds. One direction of pull shall be that tending to rotate the terminal or lead.

4.2.2.2 Bend. - Any terminal that shows visible bending in the pull test (see 4.2.2.1) shall be bent back and forth five times to an angle of 45° each side of center (total 90°).

4.2.2.3 Twist. - Stud-type terminals shall be subjected to the torque listed below:

Screw size	Torque (pound-inches)
4-40	6.5
6-40	7.6
6-32	9.0
8-36	11.5
8-32	13.5
10-32	16.0
10-24	17.5
1/4-28	30.0
1/4-20	33.0

4.2.3 Speed-torque characteristics. - The speed-torque tests shall be performed with the shunt-field excitation held constant throughout. Motors provided with means for varying shunt-field excitation shall be so adjusted that, with full-load torque applied, full-load speed is reached. Universal motors shall be tested on both alternating and direct current. The a-c frequency shall be specified on the complementary documents.

4.2.3.1 Locked rotor. - The locked rotor test shall be performed at rated voltage using preferably the cord-and-pulley method to measure minimum locked rotor torque and maximum locked-armature current. Measurements shall be made in both directions of rotation if a reversible motor is specified. (See 3.28). Measurements should be made as quickly as possible during this test to avoid overheating of the motor.

4.2.3.1.1 Motors designed to start at rated voltage. - Motors designed to start at rated voltage (without external reduced-voltage starting devices) shall be tested with

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rated voltage applied. (See 3.28). Measurements of locked-rotor current and locked-rotor torque shall be made.

4.2.3.1.2 Motors designed to start at reduced voltage. - Motors designed to start at reduced voltage (with external reduced-voltage starting devices) shall be tested with the shunt field separately excited at rated voltage. (See 3.28). Resistance shall be inserted in series with the rotor, and shall be of such magnitude as to limit the locked-rotor current to twice the full-load current. (See 3.28). Measurements of locked-rotor current and locked-rotor shall be made.

4.2.3.2 Full load. - The full-load test shall be performed at maximum ambient temperature (see 3.28) during preproduction tests but may be performed at standard test conditions (see 4.2) during inspection tests.

4.2.3.2.1 Step 1. - Full-load torque and rated voltage (see 3.28) shall be applied to the motor. Before the motor temperature has increased appreciably, the full-load speed (cold) shall be measured.

4.2.3.2.2 Step 2. - The motor shall be operated at full-load torque (see 3.28) until temperature equilibrium is reached, after which measurements of full-load current and full-load speed shall be made.

4.2.3.3 Efficiency. - The efficiency based on full-load speed shall be computed by using the following formula:

$$\text{Efficiency} = \frac{\text{Torque (oz-in, oz-ft, or ft-lb) x Speed (r.p.m.)}}{\text{K x Rated voltage (volts) x Full-load current (amps)}}$$

Where: K = 7.04 when torque is in ft-lb, or
K = 112.7 when torque is in oz-ft, or
K = 1352 when torque is in oz-in.

4.2.3.4 Change in full-load speed due to heating. - The change in full-load speed due to heating shall be computed by using the following formula:

$$\text{Percent change in full-load speed due to heating} = 100\ \frac{(\text{Full-load speed - cold}) - (\text{Full-load speed - hot})}{(\text{Full-load speed - hot})}$$

4.2.3.5 No load. - The no-load test shall be performed at maximum ambient temperature during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests. The motor shall be operated at rated voltage (see 3.28) and no load, and the no-load speed and no-load current shall be measured. No-load tests shall not be performed on series-wound and on universal motors.

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4.2.3.6 Effect of variation in line voltage. - This test shall be performed with the motor operating at maximum ambient temperature, rated voltage, and full-load torque. (See 3.28). The motor shall be operated until temperature equilibrium is reached, after which the full-load speed shall be measured. The voltage shall then be increased to 110 percent of rated voltage (See 3.28) and the speed shall be measured. The voltage shall then be decreased to 90 percent of rated voltage and the speed shall be measured. The variation from measured full-load speed for each of these conditions shall be computed in percent.

4.2.4 Locked-rotor protection (applicable only to motors for which an over-load protection device is specified.) - The locked-rotor protection test shall be performed at maximum ambient temperature (see 3.28) during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests. Rated voltage (see 3.28) shall be applied to the motor for a period of 5 hours with the rotor locked against rotation, immediately after which the no-load speed shall be measured. (See 3.12).

4.2.5 Overspeed (applicable only to motors for which external field connections are specified. - The overspeed test shall be performed at maximum ambient temperature during preproduction tests but may be performed at standard test conditions (see 4.1.2) during inspection tests. The test shall be performed at a speed not less than 125 percent of the specified no-load speed for a period of 5 minutes. This increase in speed shall be accomplished at rated voltage and no-load by inserting external resista in the field circuit. After this test the no-load speed shall be measured. (See 3.13).

4.2.6 Insulation resistance. – Measurements shall be made at 500 volts D.C. The test voltage shall be applied between ground (frame) and each winding which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. Windings which are normally connected together in service shall be considered as one winding for the purpose of this test. (See 3.14).

4.2.7 Dielectric stength. - The test voltage shall be of commercial-line frequency and of the magnitude shown in table VII. The test voltage shall be applied for 1 minute, except that for inspections tests, 120 percent of the test voltage may be applied for 1 second. Any capacitors connected to the circuit under test shall be disconnected so that they will not be stressed during test. The test voltage shall be applied between ground and each winding which is not permanently grounded by soldering, brazing, or welding, with all other windings grounded. For this test, brushes may be removed and the machine may be disassembled sufficiently to unground each winding in turn. (See 3.15).

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Table VII - Dielectric test voltages

Rated motor voltage 1/ Volts	Test voltage (r.m.s.) Volts
Less than 35 volts	500
35 to 249 volts, incl.	900
250 volts or greater	2 x rated voltage plus 1,000

1/ Motors 1/2 horsepower and larger shall be tested at 2 x rated voltage plus 1,000.

4.2.8 Concentricity. - The rotor shall be rotated slowly and examined for rubbing and binding.

4.2.9 Dynamic balance. - This test shall be performed on the completely assembled motor with all special attachments such as brakes and overspeed switches in place. The motor shall be elastically mounted so that the weight of the motor when not rotating will cause the mounting to be compressed to at least the amount shown below:

Speed (r.p.m.)	Minimum compression (inch)
900	1
1,200	9/16
1,500	3/8
1,800	1/4
3,600	1/16
7,200	1/64

For speed other than those shown above, the minimum compression, which is inversely proportional to the square of the speed, shall be interpolated. The static compression shall in no case be greater than 50 percent of the original height of the mounting. With the motor mounted, rated voltage shall be applied so that the motor operates at the specified no-load speed. (See 3.28). The motor shall be balanced with a test key in the keyway. The test key shall be a key of standard length, cut lengthwise so that when it is placed in the keyway the top of the key is flush with the top of the keyway. The amplitude of vibration shall be measured on the bearing housing with a vibration indicator, in the direction giving maximum amplitude. (See 3.17).

4.2.10 Breakdown. - The breakdown test shall be performed at rated frequency and rated voltage. The minimum breakdown torque shall be determined by any suitable means, satisfactory to the Government.

4.2.11 Temperature rise. - This test shall be performed at maximum ambient temperature for preproduction tests but may be performed at standard test conditions (see 4.1.12).

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4.2.11.1 Continuous motors. — The temperature rise of the armature and windings shall be determined by resistance measurements in an area free of drafts, with the motor mounted by normal means on a surface of low thermal conductivity such as wood. Two segments on the commutator shall be marked for identification. The resistance (r) between these marked segments shall be determined with the brushes removed. Care should be taken not to include any unnecessary external contact resistance. The motor shall then be operated at full-load torque until the temperature of the frame becomes stabilized. The temperature of the frame, measured on the external surface of the yoke at the center of the pole-shoes, shall be determined by a thermocouple or thermometer. The resistance of the field and armature between the marked segments shall be measured at 20-second intervals immediately after the motor has been stopped and shall be extrapolated back to the resistance at the time of shutdown for determination of R. The temperature rise of the field and armature winding shall be computed by the following formula:

$$\Delta T = \frac{R-r}{r}(t + 234.5) - (T-t)$$

Where:

ΔT = temperature rise, in degrees centrigrade.

R = hot resistance at time of shutdown, in ohms.

r = cold resistance, in ohms.

t = temperature of motor at time of measurement of r, in degree centigrade.

T = mean ambient temperature during final quarter of test, in degrees centigrade.

The ambient temperature shall not vary by more than 2°C. during the final quarter of this test.

4.2.11.2 Intermittent motors. — The temperature-rise for intermittent motors shall be performed in the same manner as for continuous motors except that the applicable duty cycle shall be employed. The temperature of the frame shall be considered stabilized when the temperature peaks caused by the "on" portions of the cycle reach a maximum value which is repeated for a series of consecutive cycles.

4.2.12 Temperature cycling (see 3.18). — Motors shall be subjected to the temperature-cycling test specified in method 102 of Standard MIL-STD-202. The following details and exceptions shall apply:

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(a) Test conditioner letter - A.

(b) Initial measurements - Not Applicable.

(c) Final measurements - During the last cycle, immediately after removal from the test chamber following step 1, the motor shall be subjected to the dielectric-strength test for a period of 5 seconds at 90 percent of the test voltage specified in 4.27.

4.2.13 Vibration (see 3.19). — Motors shall be subjected to the vibration test specified in method 201 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting - By normal mounting means except that shock mounts shall not be used unless they are an integral part of the motor.

(b) Initial measurements - Not Applicable.

(c) Motion - The motor shall be vibrated for a total of 7-1/2 hours, 2-1/2 hours in each of three mutually perpendicular directions. During the first 1-1/4 hours of vibration in each of these directions, the motor shall not be operated. During the remaining period of vibration in each of these directions, the motor shall be operated in accordance with the applicable duty cycle at the specified full-load speed and no-load. Full-load speed shall be obtained by adjustment of the voltage.

(d) Final measurements - Not Applicable.

4.2.14 Radio-frequency interference. — The motor shall be tested as specified, in accordance with Specification MIL-S-11748.

4.2.15 Overload. — The overload test shall be performed at maximum ambient temperature for preproduction tests but may be performed at standard test conditions (see 4.1.2) for inspection tests. The motor shall be operated at rated voltage with 125 percent of full-load torque applied for a period of 2 hours. After this test, the no-load speed shall be measured and the dielectric strength test shall be performed at 90 percent of the test voltage specified in 4.2.7. (See 3.20).

4.2.16 Low-temperature storage and operation. — The motor shall be stored for a period of 48 hours in a test chamber maintained at a temperature of $-65^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$ C.

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The temperature of the test chamber shall then be raised to -55° $^{+0}_{-3}$° C. and maintained for 24 hours. At the conclusion of this exposure period and while at this temperature, the motor shall be started with full-load torque applied at rated voltage, and the starting time needed to reach 90 percent of specified full-load speed shall be recorded. Continuous motors shall operate for 1/2 hour, and intermittent motors shall operate at the applicable duty cycle for 1/2 hour of elapsed time or six cycles, whichever is longer. (See 3.21).

4.2.17 Mechanical shock. — Motors shall be tested in accordance with test I or II, as specified. (See 3.22).

4.2.17.1 Shock test I. — The motor shall be operated at specified full-load speed and no-load. Full-load speed shall be obtained by adjustment of the voltage.

4.2.17.1.1 Equipment and mounting. — The shock-testing machine shall be that specified for lightweight equipment in Specification MIL-S-901. The motor shall be firmly secured in its normal mounting position to the mounting plate of the shock-testing machine. All shock-absorbing devices used on the motor shall be in place and unblocked. No shock mounts shall be used unless they are an integral part of the motor.

4.2.17.1.2 Number of blows. — Three blows shall be applied parallel to each of the three principal axis of the motor for a total of nine blows. The height of the hammer drop for each axial blow and each side blow shall be successively 1 foot, 2 feet, and 3 feet. The height of the hammer drop for each top blow shall be successively 2 feet, 3 feet, and 4 feet. (See 3.22.1).

4.2.17.2 Shock test II. — The motor shall be tested in accordance with specification 7201. An acceleration of 30 gravitational units shall be applied in 10 milliseconds. (See 3.22.2).

4.2.18 High-altitude operation (when applicable). — The motor shall be tested in accordance with altitude test specified in Method 105 of Standard MIL-STD-202. While in the test chamber at the specified high-altitude conditions, the motor shall be operated at approximate full-load speed and no-load, at the applicable duty cycle, for the number of hours specified for high-altitude brush life. (See 3.28). Approximate full-load speed shall be obtained by adjustment of the voltage. After removal from the test chamber, the motor shall be subjected to the dielectric-strength test at 75 percent of the test voltage specified in 4.27, and step 2 of the full-load test (see 4.2.3.2.2) shall be performed.

4.2.19 Moisture resistance (see 3.23). — Motors shall be subjected to the moisture-resistance test specified in method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

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(a) Mounting - On racks by normal mounting means.

(b) Initial measurements - not applicable.

(c) Polarization and load - not applicable.

(d) Measurements. At the completion of the test and within one hour after removal of the motor from the chamber, the insulation resistance shall be measured. The motor shall be then allowed to dry for a period of 4 to 6 hours at a temperature of 25°C ±5 C, and a relative humidity of 40 to 60 percent in order to remove surface moisture. After this drying period, the motor shall be subjected to a dielectric strength test at 90 percent of the test voltage specified. The motor shall then be operated for one hour at full-load torque, after which the full-load speed shall be measured. The insulation resistance shall again be measured after the motor has been allowed to dry for at least 48 hours at a temperature of approximately 25°C ± 5 and a relative humidity of 40 to 60 percent.

4.2.20 Tropicalization. — The motor shall be tested in accordance with fungus resistance test procedure I of Specification MIL-E-5272. At the conclusion of the test, the no-load speed shall be measured at standard test conditions (see 4.1.2), and the dielectric-strength test shall be performed at 75 percent of the test voltage specified in 4.2.7.

4.2.21 Life. — During the life test, precautions shall be taken against undue air contamination of a type not to be expected in normal service use. Reversible motors shall be operated in both directions for an approximately equal number of hours.

4.2.21.1 Continuous motors. — The life test shall be performed for a total of 2500 elapsed hours or as otherwise specified on the complementary documents. The motor shall be operated in normal operating position at rated voltage, full-load torque, and maximum ambient temperature. (See 3.28). During the first 100 elapsed hours, a switching cycle of 2 minutes on and 1 minute off shall be employed. For the remaining 900 hours, the motor shall be operated continuously except that brushes shall be replaced after the specified sea-level brush life (see 3.28), at which time brush dust shall be blown out. After the life test, the motor shall be subjected to the dielectric-strength test at 75 percent of the test voltage specified in 4.2.7, to the dynamic-balance test specified in 4.2.9, and to step II of the full-load test (see 4.2.3.2.2) at standard test conditions (see 4.2). These measurements may also be made at any time during the test, at the discretion of the Government, when performance of the motor is questionable.

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4.2.21.2 Intermittent motors. — The life test shall be performed for a total of 1,000 hours of elapsed time with the motor operating in accordance with the applicable duty cycle at rated voltage, full-load torque, and maximum ambient temperature. (See 3.28). Measurements shall be made as specified for continuous motors. During this test, brushes shall be replaced after the specified sea-level brush life (see 3.28), at which time brush dust shall be blown out.

4.2.22 Corrosion (when applicable). — Motors shall be subjected to the salt-spray (corrosion) test specified in method 101 of Standard MIL-STD-202. (See 3.24) The following details shall apply:

(a) Test condition letter - B.

(b) Standards of acceptability - Upon completion of the test, the motor shall be subjected to standard test conditions (see 4.1.2) and the marking shall be examined for legibility.

4.2.23 Motor enclosure.

4.2.23.1 Oilproof. — During this test the motor in its normal mounting position shall be operated at rated voltage with no load, at the applicable duty cycle. Hydraulic oil conforming to Specification MIL-O-5606, maintained at 69° ±5° C., shall be dripped on the motor at a rate of at least 1/2 pint per hour and allowed to flow over the motor. This test shall be performed for a period of 48 hours. After this test the motor shall be cleaned and dried and then subjected to the dielectric-strength test at 50 percent of the test voltage specified in 4.2.7. (See 3.25.7).

4.2.23.2 Explosionproof. — The motor shall be tested in accordance with explosionproof (aeronautical) test procedure II of Specification MIL-E-5272. (See 3.25.8).

4.2.23.3 Watertight. — The motor shall be mounted in its normal position and subjected to a stream of water from a 1-inch nozzle with a velocity head of 35 feet, held at a distance of 10 feet from the motor. The stream shall be applied from each direction which is likely to cause water penetration. The duration of the test shall be 60 minutes, consisting of 20 minutes with the motor at rest, 20 minutes with the motor operating at full-load speed and no load, and the final 20 minutes with the motor at rest. After this test the motor shall be disassembled sufficiently to permit inspection for water penetration. (See 3.25.9).

4.2.23.4 Submersible. — The motor shall be submerged in sea water having a specific gravity of 1.025 under a pressure equivalent of a 50-foot head for a period of 1 hour. After this test the motor shall be disassembled sufficiently to permit inspection for water penetration. (See 3.25.10).

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4.2.23.5 Dustproof. — The motor shall be tested in accordance with sand-and-dust test procedure II of Specification MIL-E-5272. While in the test chamber the motor shall be operated for 30 minutes at no load and shall then be at rest for 90 minutes. This cycle shall be repeated four times (a total of 8 hours). After this test the motor shall be subjected to step II of the full-load test (see 4.2.3.2.2) at the standard test conditions specified in 4.1.2. (See 3.25.11).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Domestic immediate use. — Motors, Fractional Horsepower, Direct Current and Universal shall be preserved and packaged to afford protection against deterioration and damage during shipment.

5.1.2 Domestic limited storage. — Motors, Fractional Horsepower, Direct Current and Universal shall be packaged Method III, per Specification MIL-P-116 as follows: Cushion each motor by wrapping in fiberboard, corrugated, single faced (flexible) conforming to Specification LLL-F-291. Place each cushioned motor up to 5 pounds in weight within a close fitting setup box, complying with Specification JAN-P-133. Seal entire closure with Tape, Paper, Gummed, Sealing and securing conforming to Specification UU-T-111.

5.1.2.1 Motors over 5 pounds in weight. — Motors over 5 pounds in weight shall be packaged Method III, using the applicable procedures as specified in 5.1.2, except each cushioned motor shall be placed within a fiber corrugated box conforming to Specification LLL-B-631, seal entire closure with Tape, Paper, Gummed conforming to Specification UU-T-111.

5.1.3 Military overseas. — Preserve and package Motors, Fractional Horsepower, Direct Current and Universal in accordance with procedures specified for designated methods as prescribed in Specification MIL-P-116 and further described in the following paragraphs.

5.1.3.1 Exterior shafts, coupling, and all unpainted exposed metal surfaces of all motors shall be cleaned and preserved prior to placement within the unit package following procedures as specified in MIL-P-116 for C-1 cleaning and P-7 or P-10 preservatives. Wrap the preserved metal parts in greaseproof barrier material, grade A, conforming to Specification JAN-P-121, of the lightest practical weight type to resist penetration of curring edges, and secure by suitable tape.

5.1.3.2 Motors (weights up to 10 pounds). — Motors up to 10 pounds in weight shall be packaged individually Method IId, as follows: Cushion each motor by wrapping in fiberboard, corrugated, single faced (flexible conforming to Specification LLL-F-291). Place each cushioned motor within a metal container Type III provided with a key wind, rip strip opening device, conforming to Specification MIL-C-124.

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5.1.3.3 Motors (weights fractional 10 pounds up to (1 horsepower 90 pounds)). — Package each fractional motor individually weighing 10 pounds to 90 pound Method IIIb, as follows: Cushion each motor on all surfaces with cells or pads or both fabricated of corrugated fiberboard, conforming to Specification 19-10 (A or C flute acceptable) and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned motor together with the required amount of desiccant as prescribed in Specification MIL-P-116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed Kraft tape conforming to Specification UU-T-111. Blunt all corners of the box. Place the boxed motor within a water-vapor-proof barrier, qualified for Class 2 up to 25 pound and class 1 for 25 to 90 pound application in accordance with the requirements of Specification MIL-B-131. Heat seal closure. Place the water-vapor-proofed motor within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape conforming to Specification JAN-P-127.

5.2 Packing.

5.2.1 Domestic, immediate use. — Motors, Fractional Horsepower, Direct Current and Universal shall be so packed as to insure arrival at destination in satisfactory condition and in containers complying with Consolidated Freight Classification Rules at lowest rates.

5.2.2 Domestic, limited storage. — Motors, Fractional Horsepower, Direct Current and Universal shall be packed in wood cleated fiberboard, wood cleated plywood, nailed wood, or wirebound boxes, conforming to Specifications PPP-B-601, NN-B-621, or NN-B-631. The gross weight shall not exceed 200 pounds for plywood or wood boxes. Closure shall be made in accordance with the applicable box specification.

5.2.3 Military, overseas. — The Motors, Fractional Horsepower, Direct Current and Universal shall be packed in a wood cleated plywood, or nailed wood box conforming to the requirements of Specifications PPP-B-601 or JAN-P-106, respectively. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container, as specified in the appendix of the applicable box specification.

5.2.3.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment. — Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

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6. NOTES

6.1 Ordering data.

6.1.1 Motors for which specification sheets have been issued. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable specification sheet.

(c) Number of sets of spare brushes. (See 3.9.4.2).

(d) Whether packing and marking are for domestic or overseas shipment. (See Section 5).

6.1.2 Motors for which specification sheets have not been issued. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) The information required on figure 2.

(c) Number of sets of spare brushes. (See 3.9.4.2).

(d) Whether packing and marking are for domestic or overseas shipment (See Section 5).

6.1.3 Indirect shipments. — The packaging, packing, and marking specified in Section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders between the manufacturer and prime contractor.

6.2 Data need to establish specification sheets. — The information listed in item (b) of 6.1.2 will be the basis for establishing an individual specification sheet. Any additional information (i.e., test data) should also be included with the request for establishment of a specification sheet.

6.3 Design and application of motors. — Matters relating to the design and application of motors covered by this specification should be referred to the bureau or service whose equipment is involved.

6.4 Sampling procedure. — The sampling procedure specified for group B inspection tests is identified for reference purposes as Plan B-2.

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6.5 Insulation materials.

6.5.1 Class A. — Class A insulation consists of:

(a) Cotton, silk, paper, and similar organic materials, either impregnated with or immersed in a liquid dielectric.

(b) Molded or laminated materials with cellulose filler, phenolic resin, or other resins of similar properties.

(c) Films and sheets of cellulose acetate and other cellulose derivatives of similar properties.

(d) Varnish (enamel) as applied to conductors.

6.5.2 Class B. — Class B insulation consists of mica, asbestos, fiberglass, and similar inorganic materials in built-up form with organic binding substances. A small proportion of class A materials may be used for structural purposes only. The electrical and mechanical properties of the insulated winding should not be impaired by application of the temperatures for class B materials.

6.5.3 Class H. — Class H insulation consists of:

(a) Mica, asbestos, fiberglass, and similar inorganic materials in built-up form with binding substances composed of silicone compounds or materials with equivalent properties.

(b) Silicone compounds in rubbery or resinous forms, or materials with equivalent properties.

A minute proportion of Class A materials may be used in class H materials only when essential for structural purposes during manufacture.

6.6 Ratings. — In order to accomplish standardization of motors for Military applications, motors should conform to the following ratings, whenever possible:

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Table VIII — Maximum ambient temperatures and maximum temperature rises (Continuous motors only)

Class of Insulation 1/	Maximum Ambient Temperature	Maximum Temperature Rise	
		Open or Protected Motor	Other Enclosure
	°C.	°C.	°C.
A	40	60	65
A	50	50	55
B	40	80	85
B	50	70	75
B	85	35	40

1/ Class H insulation is generally used in special applications and should be specified on an individual basis.

Horsepower rating		Full-load-speed ratings (r.p.m.)
0.001	1/20	850
.0015	1/12	1,140
.002	1/8	1,725
.003	1/6	3,450
.005	1/4	
.0075	1/3	
.010	1/2	
.015	3/4	
.025	1	
.035		

6.7 Frame sizes. — In order to accomplish standardization of motors for military applications, motors should conform to the frame sizes indicated in figure 3 wherever possible.

Table IX - Duty cycles

Motor Designation	Full-load	Rest
Intermittent A	5 minutes	15 minutes
Intermittent B	1 minute	

6.8 Contractual requirements. — It is recommended that the following be included in the contract.

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6.8.1 Additional inspection. — It should be understood that additional examination, measurement, and nondestructive testing of equipment may be performed by the Government when considered necessary to determine compliance with this specification and other applicable documents. Accordingly, the Government inspector may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection performed either at a Government laboratory or the contractor's plant.

6.8.2 Waiver of preproduction samples . — When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should be made by the prime contractor and should include a statement indicating that: the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order and will be at the same plant; and current production is in strict compliance with specified requirements for the new equipment except for such deviations as may be authorized on the current order. These deviations should be described in the statement. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

IDENTIFICATION FORM FOR MOTOR, DIRECT CURRENT,
(FRACTIONAL HORSEPOWER, SHUNT AND COMPOUND WOUND)

1. Motor identification:_____.
2. Type of winding:_____.
3. Horsepower rating:_____horsepower.
4. Duty cycle:_____.
5. Maximum ambient temperature:_____°C.

Figure 2. Motor-identification form

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6. Speed-torque:

Rated voltage:_____ volts.

Rated frequency (universal motors only): DC and_____cycles.

Full-load torque:_____ounce-foot.

Locked-armature current:_____ amperes, maximum.

Locked-armature torque:_____ ounce-foot, minimum.

Full-load current:_____ amperes, maximum.

Full-load speed:_____ r.p.m. ± _____ percent.

Change in full-load speed due to heating: ±_____ percent.

Efficiency:_____ percent, minimum.

No-load current:_____ amperes, maximum.

No-load speed:_____ r.p.m. ±_____ percent.

Effect of variation in line voltage on full-load speed:

percent, maximum.

7. Type(s) of enclosure:_____.

8. Direction of rotation:_____.

9. Type and length of electrical connectors:_____.

Pull-test requirement (if applicable):_____ pounds.

10. Relubrication (required or not required):_____.

11. Type of bearings:_____.

12. Free end-play and clearance:_____.

13. Weight:_____ pounds, maximum.

Figure 2. Motor-identification form (Continued)

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14. Mechanical-coupling means:_____.

15. Motor-life expectancy:_____years or hours.

Life test required_____hours at _____°C.

16. Overload protection (yes or no):_____.

17. Externally-connected field (yes or no):_____.

18. Operating position:_____.

19. Maximum temperature rise (not applicable to short-life motors):

Armature:_____°C.

Frame:_____°C.

20. Low-temperature starting time:_____.

21. Low-temperature operating time:_____.

22. Mechanical-shock test (Test I or II):_____.

23. High-altitude operation (applicable or not applicable):_____.

24. Service use of motor (air, ground, or shipboard):_____.

High-altitude brush life (at 50,000 feet):_____hours.

Sea-level brush life (at sea level):_____hours.

25. Corrosion test (applicable or not applicable):_____.

26. Class of insulation:_____.

27. Radio-frequency-interference requirements:_____.

28. Frame, mounting, electrical connector and mechanical-coupling or keyway diagram (sketch, two views). Show envelope dimensions as maximums. Show mounting and mechanical-coupling to keyway dimensions with tolerances. Show only such details as will affect interchangeability.

Figure 2. Motor-identification form (Continued)

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29. Brushes (show other features which affect interchangeability).

Brush material

Spring condition #1	inches		ounce, minimum
Spring condition #2	inches	ounce, minimum	ounce, maximum

L_s - Shunt-cable length

Shunt-cable description

	Dimension	Tolerance
L - Brush length		
W - Brush width		
T - Brush thickness		
L_n - Brush-head length		
R - Curvature radius		
D_o - Spring outside diameter		
W_1 - Shunt-terminal width		
W_e - Shunt-terminal ear width		
D_1 - Shunt-terminal diameter		

Figure 2. Motor-identification form (Continued)

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Brush material

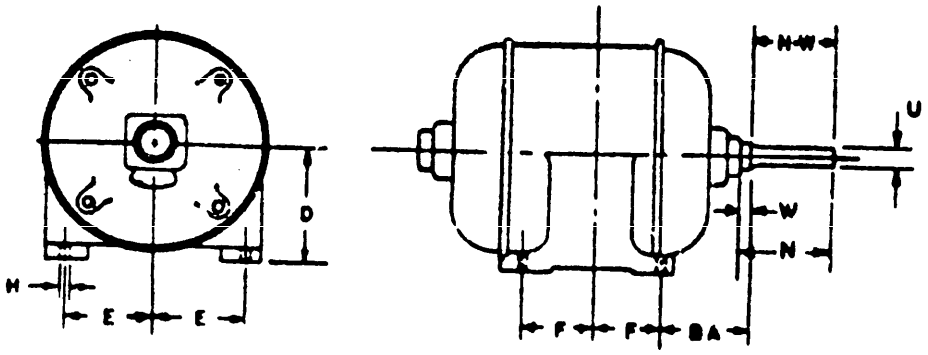
Shunt description

	Dimension	Tolerance
L - Brush length		
L_1 - Brush length		
W - Brush width		
T - Brush thickness		
R - Curvature radius		
S - Shunt length		
X - Shunt location		
Y - Shunt location		

Shunt-terminal description

Figure 2. Motor-identification form (Continued)

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Frame No.	D	E	F	BA	H	N-W*	U
42	2-5/8**	1-3/4	27/32	2-1/16	9/32 slot	1-1/8	3/8
48	3**	2-1/8	1-3/8	2-1/2	9/32 slot	1-1/2	1/2
56	3-1/2**	2-7/16	1-1/2	2-3/4	11/32 slot	1-7/8	5/8
66	4-1/8**	2-15/16	2-1/2	3-1/8	13/32 slot	2-1/4	3/4

	Key		
Frame No.	Wide	Thick	Long
42	3/64	Flat	
48	3/64	Flat	
56	3/16	3/16	1-3/8***
66	3/16	3/16	1-7/8***

*N-W is the length of shaft extension from shoulder to shaft end.
**Dimension D will never be greater than the above values for rigid-base motors, but it may be less so that shims are usually required for coupled or geared machines.
***Effective length of keyway.

STANDARD DIMENSIONS FOR NEMA FHP FRAME SIZES
(From NEMA Standard MG1-3.2)

Figure 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*;

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15415

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